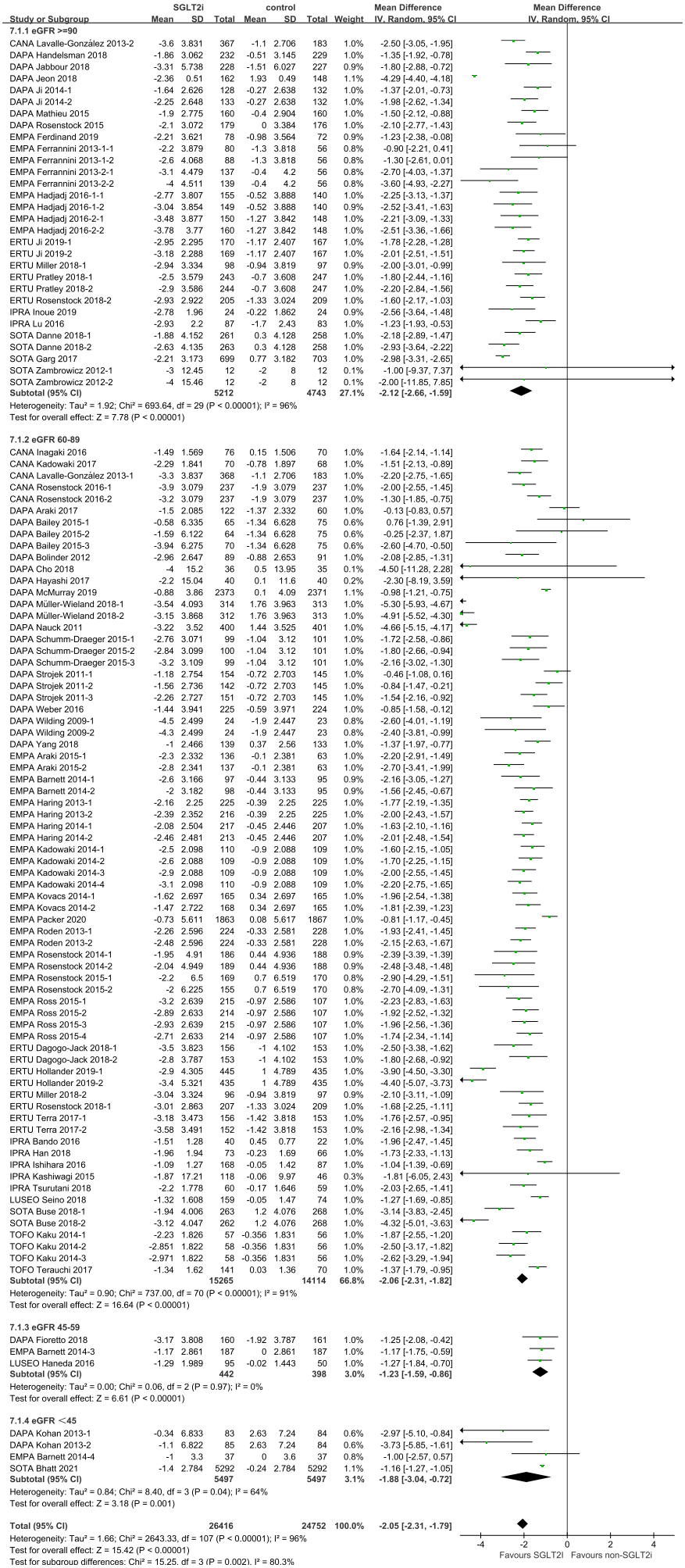

SGLT2i
control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
7.1.1 eGFR >=90
CANA Lavalle-González 2013-2 -3.6 3.831 367 -1.1 2.706 183 1.0% -2.50 [-3.05, -1.95]
DAPA Handelsman 2018 -1.86 3.062 232 -0.51 3.145 229 1.0% -1.35 [-1.92, -0.78]
DAPA Jabbour 2018 -3.31 5.738 228 -1.51 6.027 227 0.9% -1.80 [-2.88, -0.72]
DAPA Jeon 2018 -2.36 0.51 162 1.93 0.49 148 1.1% -4.29 [-4.40, -4.18]
DAPA Ji 2014-1 -1.64 2.626 128 -0.27 2.638 132 1.0% -1.37 [-2.01, -0.73]
DAPA Ji 2014-2 -2.25 2.648 133 -0.27 2.638 132 1.0% -1.98 [-2.62, -1.34]
DAPA Mathieu 2015 -1.9 2.775 160 -0.4 2.904 160 1.0% -1.50 [-2.12, -0.88]
DAPA Rosenstock 2015 -2.1 3.072 179 0 3.384 176 1.0% -2.10 [-2.77, -1.43]
EMPA Ferdinand 2019 -2.21 3.621 78 -0.98 3.564 72 0.9% -1.23 [-2.38, -0.08]
EMPA Ferrannini 2013-1-1 -2.2 3.879 80 -1.3 3.818 56 0.8% -0.90 [-2.21, 0.41]
EMPA Ferrannini 2013-1-2 -2.6 4.068 88 -1.3 3.818 56 0.8% -1.30 [-2.61, 0.01]
EMPA Ferrannini 2013-2-1 -3.1 4.479 137 -0.4 4.2 56 0.8% -2.70 [-4.03, -1.37]
EMPA Ferrannini 2013-2-2 -4 4.511 139 -0.4 4.2 56 0.8% -3.60 [-4.93, -2.27]
EMPA Hadjadj 2016-1-1 -2.77 3.807 155 -0.52 3.888 140 1.0% -2.25 [-3.13, -1.37]
EMPA Hadjadj 2016-1-2 -3.04 3.854 149 -0.52 3.888 140 0.9% -2.52 [-3.41, -1.63]
EMPA Hadjadj 2016-2-1 -3.48 3.877 150 -1.27 3.842 148 1.0% -2.21 [-3.09, -1.33]
EMPA Hadjadj 2016-2-2 -3.78 3.77 160 -1.27 3.842 148 1.0% -2.51 [-3.36, -1.66]
ERTU Ji 2019-1 -2.95 2.295 170 -1.17 2.407 167 1.0% -1.78 [-2.28, -1.28]
ERTU Ji 2019-2 -3.18 2.288 169 -1.17 2.407 167 1.0% -2.01 [-2.51, -1.51]
ERTU Miller 2018-1 -2.94 3.334 98 -0.94 3.819 97 0.9% -2.00 [-3.01, -0.99]
ERTU Pratley 2018-1 -2.5 3.579 243 -0.7 3.608 247 1.0% -1.80 [-2.44, -1.16]
ERTU Pratley 2018-2 -2.9 3.586 244 -0.7 3.608 247 1.0% -2.20 [-2.84, -1.56]
ERTU Rosenstock 2018-2 -2.93 2.922 205 -1.33 3.024 209 1.0% -1.60 [-2.17, -1.03]
IPRA Inoue 2019 -2.78 1.96 24 -0.22 1.862 24 0.9% -2.56 [-3.64, -1.48]
IPRA Lu 2016 -2.93 2.2 87 -1.7 2.43 83 1.0% -1.23 [-1.93, -0.53]
SOTA Danne 2018-1 -1.88 4.152 261 0.3 4.128 258 1.0% -2.18 [-2.89, -1.47]
SOTA Danne 2018-2 -2.63 4.135 263 0.3 4.128 258 1.0% -2.93 [-3.64, -2.22]
SOTA Garg 2017 -2.21 3.173 699 0.77 3.182 703 1.0% -2.98 [-3.31, -2.65]
SOTA Zambrowicz 2012-1 -3 12.45 12 -2 8 12 0.1% -1.00 [-9.37, 7.37]
SOTA Zambrowicz 2012-2 -4 15.46 12 -2 8 12 0.1% -2.00 [-11.85, 7.85]
Subtotal (95% CI) 5212 4743 27.1% -2.12 [-2.66, -1.59]
Heterogeneity: Tau² = 1.92; Chi² = 693.64, df = 29 (P < 0.00001); I² = 96%
Test for overall effect: Z = 7.78 (P < 0.00001)
7.1.2 eGFR 60-89
CANA Inagaki 2016 -1.49 1.569 76 0.15 1.506 70 1.0% -1.64 [-2.14, -1.14]
CANA Kadowaki 2017 -2.29 1.841 70 -0.78 1.897 68 1.0% -1.51 [-2.13, -0.89]
CANA Lavalle-González 2013-1 -3.3 3.837 368 -1.1 2.706 183 1.0% -2.20 [-2.75, -1.65]
CANA Rosenstock 2016-1 -3.9 3.079 237 -1.9 3.079 237 1.0% -2.00 [-2.55, -1.45]
CANA Rosenstock 2016-2 -3.2 3.079 237 -1.9 3.079 237 1.0% -1.30 [-1.85, -0.75]
DAPA Araki 2017 -1.5 2.085 122 -1.37 2.332 60 1.0% -0.13 [-0.83, 0.57]
DAPA Bailey 2015-1 -0.58 6.335 65 -1.34 6.628 75 0.6% 0.76 [-1.39, 2.91]
DAPA Bailey 2015-2 -1.59 6.122 64 -1.34 6.628 75 0.6% -0.25 [-2.37, 1.87]
DAPA Bailey 2015-3 -3.94 6.275 70 -1.34 6.628 75 0.6% -2.60 [-4.70, -0.50]
DAPA Bolinder 2012 -2.96 2.647 89 -0.88 2.653 91 1.0% -2.08 [-2.85, -1.31]
DAPA Cho 2018 -4 15.2 36 0.5 13.95 35 0.1% -4.50 [-11.28, 2.28]
DAPA Hayashi 2017 -2.2 15.04 40 0.1 11.6 40 0.2% -2.30 [-8.19, 3.59]
DAPA McMurray 2019 -0.88 3.86 2373 0.1 4.09 2371 1.1% -0.98 [-1.21, -0.75]
DAPA Müller-Wieland 2018-1 -3.54 4.093 314 1.76 3.963 313 1.0% -5.30 [-5.93, -4.67]
DAPA Müller-Wieland 2018-2 -3.15 3.868 312 1.76 3.963 313 1.0% -4.91 [-5.52, -4.30]
DAPA Nauck 2011 -3.22 3.52 400 1.44 3.525 401 1.0% -4.66 [-5.15, -4.17]
DAPA Schumm-Draeger 2015-1 -2.76 3.071 99 -1.04 3.12 101 1.0% -1.72 [-2.58, -0.86]
DAPA Schumm-Draeger 2015-2 -2.84 3.099 100 -1.04 3.12 101 1.0% -1.80 [-2.66, -0.94]
DAPA Schumm-Draeger 2015-3 -3.2 3.109 99 -1.04 3.12 101 1.0% -2.16 [-3.02, -1.30]
DAPA Strojek 2011-1 -1.18 2.754 154 -0.72 2.703 145 1.0% -0.46 [-1.08, 0.16]
DAPA Strojek 2011-2 -1.56 2.736 142 -0.72 2.703 145 1.0% -0.84 [-1.47, -0.21]
DAPA Strojek 2011-3 -2.26 2.727 151 -0.72 2.703 145 1.0% -1.54 [-2.16, -0.92]
DAPA Weber 2016 -1.44 3.941 225 -0.59 3.971 224 1.0% -0.85 [-1.58, -0.12]
DAPA Wilding 2009-1 -4.5 2.499 24 -1.9 2.447 23 0.8% -2.60 [-4.01, -1.19]
DAPA Wilding 2009-2 -4.3 2.499 24 -1.9 2.447 23 0.8% -2.40 [-3.81, -0.99]
DAPA Yang 2018 -1 2.466 139 0.37 2.56 133 1.0% -1.37 [-1.97, -0.77]
EMPA Araki 2015-1 -2.3 2.332 136 -0.1 2.381 63 1.0% -2.20 [-2.91, -1.49]
EMPA Araki 2015-2 -2.8 2.341 137 -0.1 2.381 63 1.0% -2.70 [-3.41, -1.99]
EMPA Barnett 2014-1 -2.6 3.166 97 -0.44 3.133 95 0.9% -2.16 [-3.05, -1.27]
EMPA Barnett 2014-2 -2 3.182 98 -0.44 3.133 95 0.9% -1.56 [-2.45, -0.67]
EMPA Haring 2013-1 -2.16 2.25 225 -0.39 2.25 225 1.0% -1.77 [-2.19, -1.35]
EMPA Haring 2013-2 -2.39 2.352 216 -0.39 2.25 225 1.0% -2.00 [-2.43, -1.57]
EMPA Haring 2014-1 -2.08 2.504 217 -0.45 2.446 207 1.0% -1.63 [-2.10, -1.16]
EMPA Haring 2014-2 -2.46 2.481 213 -0.45 2.446 207 1.0% -2.01 [-2.48, -1.54]
EMPA Kadowaki 2014-1 -2.5 2.098 110 -0.9 2.088 109 1.0% -1.60 [-2.15, -1.05]
EMPA Kadowaki 2014-2 -2.6 2.088 109 -0.9 2.088 109 1.0% -1.70 [-2.25, -1.15]
EMPA Kadowaki 2014-3 -2.9 2.088 109 -0.9 2.088 109 1.0% -2.00 [-2.55, -1.45]
EMPA Kadowaki 2014-4 -3.1 2.098 110 -0.9 2.088 109 1.0% -2.20 [-2.75, -1.65]
EMPA Kovacs 2014-1 -1.62 2.697 165 0.34 2.697 165 1.0% -1.96 [-2.54, -1.38]
EMPA Kovacs 2014-2 -1.47 2.722 168 0.34 2.697 165 1.0% -1.81 [-2.39, -1.23]
EMPA Packer 2020 -0.73 5.611 1863 0.08 5.617 1867 1.0% -0.81 [-1.17, -0.45]
EMPA Roden 2013-1 -2.26 2.596 224 -0.33 2.581 228 1.0% -1.93 [-2.41, -1.45]
EMPA Roden 2013-2 -2.48 2.596 224 -0.33 2.581 228 1.0% -2.15 [-2.63, -1.67]
EMPA Rosenstock 2014-1 -1.95 4.91 186 0.44 4.936 188 0.9% -2.39 [-3.39, -1.39]
EMPA Rosenstock 2014-2 -2.04 4.949 189 0.44 4.936 188 0.9% -2.48 [-3.48, -1.48]
EMPA Rosenstock 2015-1 -2.2 6.5 169 0.7 6.519 170 0.8% -2.90 [-4.29, -1.51]
EMPA Rosenstock 2015-2 -2 6.225 155 0.7 6.519 170 0.8% -2.70 [-4.09, -1.31]
EMPA Ross 2015-1 -3.2 2.639 215 -0.97 2.586 107 1.0% -2.23 [-2.83, -1.63]
EMPA Ross 2015-2 -2.89 2.633 214 -0.97 2.586 107 1.0% -1.92 [-2.52, -1.32]
EMPA Ross 2015-3 -2.93 2.639 215 -0.97 2.586 107 1.0% -1.96 [-2.56, -1.36]
EMPA Ross 2015-4 -2.71 2.633 214 -0.97 2.586 107 1.0% -1.74 [-2.34, -1.14]
ERTU Dagogo-Jack 2018-1 -3.5 3.823 156 -1 4.102 153 1.0% -2.50 [-3.38, -1.62]
ERTU Dagogo-Jack 2018-2 -2.8 3.787 153 -1 4.102 153 1.0% -1.80 [-2.68, -0.92]
ERTU Hollander 2019-1 -2.9 4.305 445 1 4.789 435 1.0% -3.90 [-4.50, -3.30]
ERTU Hollander 2019-2 -3.4 5.321 435 1 4.789 435 1.0% -4.40 [-5.07, -3.73]
ERTU Miller 2018-2 -3.04 3.324 96 -0.94 3.819 97 0.9% -2.10 [-3.11, -1.09]
ERTU Rosenstock 2018-1 -3.01 2.863 207 -1.33 3.024 209 1.0% -1.68 [-2.25, -1.11]
ERTU Terra 2017-1 -3.18 3.473 156 -1.42 3.818 153 1.0% -1.76 [-2.57, -0.95]
ERTU Terra 2017-2 -3.58 3.491 152 -1.42 3.818 153 1.0% -2.16 [-2.98, -1.34]
IPRA Bando 2016 -1.51 1.28 40 0.45 0.77 22 1.0% -1.96 [-2.47, -1.45]
IPRA Han 2018 -1.96 1.94 73 -0.23 1.69 66 1.0% -1.73 [-2.33, -1.13]
IPRA Ishihara 2016 -1.09 1.27 168 -0.05 1.42 87 1.0% -1.04 [-1.39, -0.69]
IPRA Kashiwagi 2015 -1.87 17.21 118 -0.06 9.97 46 0.3% -1.81 [-6.05, 2.43]
IPRA Tsurutani 2018 -2.2 1.778 60 -0.17 1.646 59 1.0% -2.03 [-2.65, -1.41]
LUSEO Seino 2018 -1.32 1.608 159 -0.05 1.47 74 1.0% -1.27 [-1.69, -0.85]
SOTA Buse 2018-1 -1.94 4.006 263 1.2 4.076 268 1.0% -3.14 [-3.83, -2.45]
SOTA Buse 2018-2 -3.12 4.047 262 1.2 4.076 268 1.0% -4.32 [-5.01, -3.63]
TOFO Kaku 2014-1 -2.23 1.826 57 -0.356 1.831 56 1.0% -1.87 [-2.55, -1.20]
TOFO Kaku 2014-2 -2.851 1.822 58 -0.356 1.831 56 1.0% -2.50 [-3.17, -1.82]
TOFO Kaku 2014-3 -2.971 1.822 58 -0.356 1.831 56 1.0% -2.62 [-3.29, -1.94]
TOFO Terauchi 2017 -1.34 1.62 141 0.03 1.36 70 1.0% -1.37 [-1.79, -0.95]
Subtotal (95% CI) 15265 14114 66.8% -2.06 [-2.31, -1.82]
Heterogeneity: Tau² = 0.90; Chi² = 737.00, df = 70 (P < 0.00001); I² = 91%
Test for overall effect: Z = 16.64 (P < 0.00001)
7.1.3 eGFR 45-59
DAPA Fioretto 2018 -3.17 3.808 160 -1.92 3.787 161 1.0% -1.25 [-2.08, -0.42]
EMPA Barnett 2014-3 -1.17 2.861 187 0 2.861 187 1.0% -1.17 [-1.75, -0.59]
LUSEO Haneda 2016 -1.29 1.989 95 -0.02 1.443 50 1.0% -1.27 [-1.84, -0.70]
Subtotal (95% CI) 442 398 3.0% -1.23 [-1.59, -0.86]
Heterogeneity: Tau² = 0.00; Chi² = 0.06, df = 2 (P = 0.97); I² = 0%
Test for overall effect: Z = 6.61 (P < 0.00001)
7.1.4 eGFR <45
DAPA Kohan 2013-1 -0.34 6.833 83 2.63 7.24 84 0.6% -2.97 [-5.10, -0.84]
DAPA Kohan 2013-2 -1.1 6.822 85 2.63 7.24 84 0.6% -3.73 [-5.85, -1.61]
EMPA Barnett 2014-4 -1 3.3 37 0 3.6 37 0.8% -1.00 [-2.57, 0.57]
SOTA Bhatt 2021 -1.4 2.784 5292 -0.24 2.784 5292 1.1% -1.16 [-1.27, -1.05]
Subtotal (95% CI) 5497 5497 3.1% -1.88 [-3.04, -0.72]
Heterogeneity: Tau² = 0.84; Chi² = 8.40, df = 3 (P = 0.04); I² = 64%
Test for overall effect: Z = 3.18 (P = 0.001)
Total (95% CI) 26416 24752 100.0% -2.05 [-2.31, -1.79]
Heterogeneity: Tau² = 1.66; Chi² = 2643.33, df = 107 (P < 0.00001); I² = 96%
Test for overall effect: Z = 15.42 (P < 0.00001)
Test for subgroup differences: Chi² = 15.25, df = 3 (P = 0.002), I² = 80.3%
-4 -2 0 2 4
Favours SGLT2i Favours non-SGLT2i